

FERRARI CECILIA

PhD in Astroparticle Physics

I am interested in **detector R&D and data analysis for Beyond Standard Model physics experiments**. I have experience in neutrino mass, neutrino oscillation, nuclear and dark matter physics in ultra-low-background experiments. I currently am a postdoc associate at MIT working in neutrino and dark matter experiments in Russell group.

EDUCATION

2020-2025

Gran Sasso Science Institute
(L'Aquila, Italy)

PhD Degree in Astroparticle Physics

cum Laude

Title: Radon-Induced Lead Isotopes in XENONnT: Studies on Branching Ratios and Surface Background

Supervisor: **A.D.Ferella**

2018-2020

Milano-Bicocca University
(Milan, Italy)

Master Degree in Experimental Particle Physics

110/110 cum Laude, weighted exams average grade: 29.8/30

Title: Characterization and Pulse Processing of TES Microcalorimeters Array for the HOLMES Experiment

Supervisor: **A.E.L.Nucciotti**

2015-2018

Milano-Bicocca University
(Milan, Italy)

Bachelor Degree in Physics

110/110 cum Laude, weighted exams average grade: 29.42/30

Title: A Time Resolution Study of a CMS timing detector

Supervisor: **T.Tabarelli de Fatis**

RESEARCH EXPERIENCE

Jun 2025-present

MIT
(Boston, Massachusetts)

Scintillating GaAs studies for sub-GeV dark matter searches

- Sensitivity studies for sub-GeV dark matter searches for a scintillating GaAs based device
- Laboratory characterization of GaAs crystal response to infrared laser

Charge Readout Expert for NDLAr DUNE prototype

- Charge readout expert in 2x2 DUNE NDLAr prototype
- Served as shifter in 2x2 Run2

Hardware and Software studies in DUNE collaboration

- Characterization studies LArPix ASIC-based charge readout system
- Simulation analyses for DUNE NDLAr requirements verification in terms of tracking and spatial resolution

MARLENA MIT LAr TPC commissioning

- Material procurement for LAr TPC construction and commissioning in the MIT Russell lab
- Assembly of a LAr molecular sieve based filter

Neutrino Phenomenology studies

- Thermal solar neutrino phenomenology studies

Jul 2021-May 2025

Laboratori Nazionali del
Gran Sasso
(Assergi, Italy)

Member of the DAREDEVIL light dark matter project

- light dark matter sensitivity studies with milli-Kelvin macrocalorimeters with GaAs semiconductor target

Co-leader of high-energy ER XENONnT analysis group

Coordination of the analysis group and developer of background modeling and fitting framework for XENONnT high-energy electron recoils analyses

Data Analysis for Nuclear Physics and WIMP searches

- ^{212}Pb and ^{214}Pb beta decay branching ratios measurement with XENONnT experiment
- Physics-driven 6D model of the surface background for XENONnT WIMP searches with flamedisx, developed in participation with LZ Collaboration

Run Coordinator of the XENONnT experiment

One month as Run Coordinator for the XENONnT Experiment at LNGS, following calibration campaigns, monitoring the status of every experimental sub-system and coordinating on-site shifters and experts operations.

Jun 2022

Stockholm University
(Stockholm, Sweden)

Visiting PhD student (Supervisors: J.Conrad and J.Mahlstedt)

- First characterization of novel ABALONE photosensor in cryostat
- Development of flamedisx software for WIMP inference studies

Dec 219 - Oct 2020

Milano-Bicocca University
(Milan, Italy)

Master Thesis in HOLMES experiment

- Characterization of TES-based microcalorimeters multiplexed array: evaluation of V-I curves and thermal conductances to the bath of 16 TESs coupled to multiplexed RF-SQUIDS NIST μmux17a chip.
- Studies for a novel pulse filtering technique to process pile-up events, resulting in a 19% improvement of HOLMES sensitivity on neutrino mass. Corresponding author of a paper and a proceeding.

Jun 2019 - Sep 2019

CERN - EP Department
(Geneva, Switzerland)

Summer Student Internship in CLICdp experiment

Long-Lived Particles detection sensitivity studies for the proposed Compact Linear Collider CLICdp experiment

Jun 2018 - Oct 2018

Milano-Bicocca University
(Milan, Italy)

Bachelor Thesis in CMS MTD Milano-Bicocca group

Characterization and noise analysis of a LYSO-coupled SiPM prototype for the CMS MTD Barrel Timing Layer

SELECTED CONFERENCES AND SEMINARS

Invited Talks and Seminars

[Mar 2024] at Princeton Physics Dep. - Dicke Fellowship Seminar:
More than Background: Lead Studies in XENONnT

[Mar 2024] at 2nd Quarkonia Meets Dark Matter (QMDM) workshop -
Talk in **Plenary** session:
Advancements in Direct Dark Matter Detection with XENONnT

Invited Talks and Seminars

[Nov 2023] at UnivAQ Thursday Morning Science seminars cycle:
Direct dark matter searches with the XENONnT experiment

Other Contributions

[Oct 2025] at CPAD - Talk in **Parallel** session:
Measuring the optical scattering in n-type GaAs that could explain its high cryogenic scintillation luminosity

[Oct 2024] at LRT - Talk in **Plenary** session:
 ^{212}Pb and ^{214}Pb Beta Decay Branching Ratios Measurement with XENONnT

[Jun 2024] at NEUTRINO - Poster contribution:
 ^{214}Pb Branching Ratios Measurement with XENONnT

[Sep 2023] at TeVPA - Talk in **Plenary** session:
XENONnT direct dark matter searches: the latest results

[Aug 2023] at TAUP - Poster contribution:
The physics-driven surface background model for XENONnT

[Apr 2023] at IFAE - Talk in **Plenary** session:
Primi risultati dell'esperimento XENONnT

[Sep 2022] at DMNet Int. Symposium - Talk in **Plenary** session:
Flamedisx. Fast Likelihood Analysis in MorE DimensionS for Xenon TPCs

[May 2022] at PM2021 - Poster contribution:
The ABALONE Photodetector

[Feb 2020] at "Det. of the Absolute Electron (anti)neutrino Mass" - Poster contribution:
Optimization of TES-based microcalorimeters for PTOLEMY

WORKSHOPS and PhD SCHOOLS

7. [Jun 2023] International School of Subnuclear Physics ISSP EMFCSC, Erice (IT)
6. [Oct 2022] The complexity of the Cosmos, L'Aquila (IT)
5. [May 2022] INFN School of Statistics 2022, Paestum (IT)
4. [Jun 2021] Quantum Computation and Sensing, online event
3. [Mar 2021] Frontiers in Nuclear and Hadronic Physics 2021, online event
2. [Jul 2019] Summer School Lectures on high energy particle physics at CERN, Geneva (CH)
1. [Sep 2018] Summer School on plasma physics and fusion research at IPP, Greifswald (DE)

ACHIEVEMENTS, SCHOLARSHIPS and AWARDS

8. [2024] Shortlisted for the MIT 2024-2027 Pappalardo Fellowship
7. [2024] Winner of the Princeton 2024-2027 Dicke Fellowship. Declined for another position.
6. [Jun 2023] "Chien-Shiung Wu" Best New Talent diploma at ISSP EMFCSC 2023
5. [Sep 2022] Contribution at the 106th SIF National Congress Italian Physical Society "awarded with the publication on Il Nuovo Cimento"
4. [2020] Winner of a GSSI PhD scholarship
3. [2020] Winner of a Milano-Bicocca University PhD scholarship
2. [2019] Winner of a CERN Summer Student scholarship
1. [2018] Selected as participant for the IPP Summer School on plasma physics

TEACHING and SUPERVISION DUTIES

- [2025-now] Thomas Barbera – Grad Student
- [2023] Andrea Melchiorre – Master Thesis: Characterization of SiPMs for dark matter re-search experiments.
- [2023] Chiara Di Donato – Master Thesis: XENONnT electronic recoil response characterization.
- [2023] Matteo Giammarco – Bachelor Thesis: Misura di coincidenze ritardate in XENONnT dalla calibrazione con sorgente di ^{222}Rn .
- [2020] 24h of tutoring frontal classes for Classical Mechanics course at Milano-Bicocca University

CONFERENCES and SEMINARS ORGANIZATION

- [2024] Organizer of three XENONnT Analysis Workshops in L'Aquila-LNGS (Mar 2024, Sep 2024, Dec 2024) and Amsterdam (Sep 2024)
- [Jul 2024] Member of the **international IDM Conference LOC** in L'Aquila
- [2023,2024] Organizer of three XENONnT Collaboration Meetings in L'Aquila University and LNGS
- [2023,2024] Organizer of two PhTea Talks interdisciplinary informal seminars cycles for PhDs

COMPUTING AND LANGUAGE SKILLS

Computing Skills

- *Operating systems*: Unix/Linux, Windows (basics)
- *Programming Languages*:
 - *Advanced*: Python, bash
 - *Intermediate*: C/C++, MATLAB
 - *Basic*: R, Pearl
- *Software*: Git (and GitHub), ROOT Data Analysis Framework, COM-SOL Multiphysics, Stan (basics), Microsoft and Office suite (ECDL Full Standard Certificate [Sep 2015])
- *Typesetting*: $\text{\LaTeX}$, Markdown

Language Skills

- Italian (Native)
- English (Fluent)
- French (B1 - BBetween Unimib Open Badge [Jan 2019])

EXTRACURRICULAR ACTIVITIES

- [Mar 2023 - present] President (three-years term) and Funder of the GSSI Alumni association
- [Dec 2023 - present] Member of the Researcher for Gender Equity GSSI Committee
- [2017-2019] Student representative in three Milano-Bicocca University Department Councils

OUTREACH ACTIVITIES

- [Ed.s 2023/2024] Organizer of "Parentesi" scientific literary symposium in L'Aquila
- [Since 2023] Trained guide for underground laboratory visits at LNGS
- [Since 2022] Member of Public Relation team of the XENONnT experiment
- [Since 2021] Organizer and public speaker in L'Aquila SHARPER researchers night
- [Since 2021] Scientific Jury Member of the Italian ASIMOV scientific literature prize